

Sinothando Mafu

20 September 2026

## Assignment 2

### The Language of Business Analytics

#### Section 1

##### 1. BUSINESS ANALYTICS

The discipline of using data not only to report on the past (as Business Intelligence does) or handle the technical side of data work (as Data Analytics does), but to ask "why" and solve problems before they become a crisis, helping businesses make better decisions (Tableau, n.d.).

##### Example:

UPS uses data from its delivery trucks to predict the fastest route for each driver every day and recommends that route to them. This cut about 100 million miles driven per year and saved the company hundreds of millions of dollars.

##### 2. DATA ANALYTICS

The process of cleaning, preparing and analyzing stored data to answer questions, identify trends, and extract insights. It isn't limited to any field and can be used in areas like science, government and business (Tableau, n.d.).

##### Example:

Johns Hopkins university built an online dashboard that collects COVID-19 case, death and recovery

numbers from around the world and shows them on a live map. This shows data analytics being used outside business, turning raw data into information that anyone could follow (Kelly, 2020).

### 3. BUSINESS INTELLIGENCE (BI)

The process of collecting, storing and analyzing data from business operations, rather than from any source, to assess how the business runs. It looks at the past and present using tools and reports, usually dashboards, that show what is happening in the business, helping it track performance and remain compliant (Tableau, n.d.).

#### Example:

Walmart's Retail Link gives suppliers reports on their product's sales and inventory, so they can track performance and reduce inventory risk (Gloy, 2024).

### 4. DATA-DRIVEN DECISION MAKING

The method of making decisions based on evidence from data, and not just on a gut feeling (Cote, 2021; Joubert, 2024).

#### Example:

Starbucks uses a mapping tool called Atlas to analyse data such as population, income, traffic and competitors before deciding where to open a new store, rather than relying on a guess (Harvard B3, n.d.).

## 5. BIG DATA

When data is large, fast and varied to the extent that normal software cannot easily handle it, unlike analytics, which examines data to find insights, big data describes the data itself (Roch et al, 2022).

### Example:

Starbucks makes around 90 million transactions a week across almost 25 000 stores, and its mobile app adds millions more orders, so it collects a huge and constant stream of data (Harvard Business School Digital Initiative, n.d.b).

## 6. DESCRIPTIVE ANALYTICS

Looks at existing data to better understand and describe exactly what happened. It does not ask why sales are low, or predict what will happen next, or prescribe what a business should do (Microsoft, 2025).

### Example:

Woolworths Holdings looked at its Financial Year 2025 results compared with FY2024 and reported that turnover and concession sales were R81.0 billion, an increase of 6.1% (Woolworths Holdings, 2025).

## 7. DIAGNOSTIC ANALYTICS

Goes further than reporting and looks at why something happened. Businesses use it to understand the causes of a result so they

can make informed decisions and address potential problems (Microsoft, 2025).

#### Example:

Woolworths Holdings explained that the Country Road Group had a decline of more than 100%, from a profit of A\$51.3m to a loss of A\$18.1m in the previous year, caused by a weak retail environment (Woolworths Holdings, 2025).

### 8. PREDICTIVE ANALYTICS

Uses past patterns to determine the probability of what happens next instead of comparing solutions and looking at why something happened (Microsoft, 2025).

#### Example:

Woolworths used their first seven weeks of trading data to estimate future sales growth for FY26, including 6,9% for Food, 8,3% for FBH and 1,2% for CRG (Woolworths Holdings, 2025).

### 9. PRESCRIPTIVE ANALYTICS

Goes beyond prediction, and looks at existing data, compares different solutions and helps businesses determine what decision(s) should be made (Microsoft, 2025).

#### Example:

Woolworths used its financial data to plan how investments across technology, stores, customer experience and supply chain were

to be allocated to support future growth. This was a recommended course of action (Woolworths Holdings, 2025).

## 10. DATA LITERACY

The skill of reading, understanding, questioning and communicating data-driven insights (SAS Institute, 2020).

### Example:

An employee at Woolworths correctly read the sales dashboard and used that understanding to spot trends and interpret charts, then explained the trends to their manager to support decision-making.

## Section 2

## 11. STRUCTURED DATA

Consistent, pre-defined format where data is rigidly fixed into rows and columns making it easy to understand and analyse. No extra effort or work is needed to handle whatever data is stored (IBM, n.d.).

### Example:

Shoprite's sales transaction table with columns such as Date, Product ID, Quantity and Price (both quantitative and qualitative data exists or is included).

## 12. UNSTRUCTURED DATA

Unorganized, inconsistent format where data is kept in its original format when it was created and extra tools are needed to make sense of it (IBM, n.d.).

Example:

Videos/Files, Music, Phonecalls, Adverts, Emails (both textual and nontextual data exists or is included).

## 13. DATA WAREHOUSE

Where data from different parts of a business can be extracted, transformed and loaded into a large central storage system for analysis (IBM, n.d.).

Example:

Google BigQuery uses ETL to pull sales, inventory, and customer data from separate systems into one place so it can all be analysed altogether.

## 14. DATA MINING

A versatile tool that can be applied to different types of data and used to discover patterns, relationships, and trends to improve business behaviour (IBM, n.d.).

Example:

Woolworths uses AI and its data analytics centre to mine customer and loyalty data, helping the company make more personalised, customer-focused decisions through initiatives

Such as its in development MyDifference programme (Woolworths Holdings, 2025).

#### 15. DATA VISUALIZATION

Data telling the story through charts, graphs and maps where information is summarized, giving key findings or highlights of data for quick decision making (IBM, n.d.).

##### Example:

Woolworths used bar charts in its FY25 Analyst Presentation to visually compare capital expenditure for 2024, 2025 and its 2026 plan, allowing shareholders to easily see spending trends at a glance (Woolworths Holdings, 2025).

#### 16. DASHBOARD

A single illustration that brings together various data visualizations (KPI's, texts, figures, and tables) to give a multidimensional view of collected data (O'Reilly Media, 2025).

##### Example:

In Tableau, a dashboard combines multiple sheets, filters and visual objects into one interactive view, allowing users to compare different metrics at the same time (Tableau, n.d.).

#### 17. METRIC

Any measurable number a business tracks to assess performance, such as cost, sales, or efficiency figures and whether or not it's tied to a specific strategic goal (Klipfolio, 2026).

### Example:

Woolworths reported a gross profit margin of 24,9% for FY25, up 0,2 percentage points on the previous year which was a number the company monitored but does not attach to a specific strategic target, unlike ROCE (Woolworths Holdings, 2025).

## 18. KPI

A key performance indicator is a metric specifically selected because it directly reflects progress toward a strategic business objective (Klipfolio, 2026).

### Example:

Woolworths uses ROCE as a KPI to measure how well it uses its capital. In FY25, its ROCE was 16,4%, compared to a WACC of 12,4%, meaning it was above its target (Woolworths Holdings, 2025).

## 19. BENCHMARK

Point of reference that a business can use to compare its performance with an industry standard, a competitor, or its own previous performance (IBM, 2026).

### Example:

BrightLearn compares its number of enrolled students each month against the previous month's numbers (a past-performance benchmark) to check whether platform growth is on track.

## 20. SEGMENTATION

The process of dividing potential customers into different groups based on their shared or distinctive characteristics (Pearson, 2020).

### Example:

Woolworths could divide its customers into different groups based on what they buy and what they are interested in, so that it can target each group with products and promotions that suit them.

## Section 3

## 21. CONVERSION RATE

A measurement of the percentage of people  $[(\text{Number of conversions} \div \text{Total visitors}) \times 100]$  who make an actual purchase from the stores they visit compared to the number of people that visited said store overall (HubSpot Inc, 2026).

### Example:

Many people visit Woolworths instore and online but not everyone makes an actual purchase, therefore the conversion rate is always calculated (Woolworths Holdings, 2025).

## 22. CUSTOMER ACQUISITION COST

The total amount a company spends on ads, sales or buying relevant resources with the objective of gaining new customers. The total cost is then calculated by dividing it by the number of new customers gained within that

specific period (Hubspot, 2026).

Example:

Hubspot spent approximately \$1.38 billion on sales and marketing in FY2025 and the customer base increased from 247,939 customers in FY2024 to 288,706 customers in FY2025. Based on said figures, Hubspot's estimated customer Acquisition Cost was approximately \$33,836 per new customer (Average, 2026).

### 23. CUSTOMER LIFETIME VALUE (CLV/LTV)

An estimate of the total profit a business generates from one customer throughout their time as a customer, rather than from one purchase (Investopedia, n.d.).

Example:

I pay Virgin Active roughly R800/month and have been a member for 3 years, giving an estimated CLV of  $R800 \times 36 \text{ months} = R28\,800$ .

### 24. CHURN RATE

The percentage of people who stop doing business with a company during a specific period (Investopedia, n.d.).

Example:

Netflix had an increase in its churn rate in the first quarter of 2022. Their active monthly churn rate reached 3.3% by the end of March 2022 due to an increase in cancellations following its January 2022 price increase (Antenna, 2022).

## 25. CUSTOMER RETENTION RATE

The opposite of churn rate; it is the percentage of customers that a business manages to keep over a specific period (Salesforce, 2026).

### Example:

Netflix had a customer retention rate of 72% after six months for customers who subscribed between January 2021 and January 2022 meaning 72% of these customers were still subscribed to Netflix six months after joining the service (Bloomberg second measure, 2022).

## 26. NET PROMOTER SCORE (NPS)

The likelihood of customers recommending a service they use, measured by a score from -100 to 100. This asks the customer "how likely are you to recommend this to a friend on a scale of 0-10?" (Focus Feedback, n.d.).

### Example:

An airline calculated its NPS by subtracting the percentage of detractors from the percentage of promoters. For example, 60% promoters and 15% detractors would give an NPS of 45 (Focus Feedback, n.d.).

## 27. Average ORDER VALUE

The average amount customers spend per order, calculated by dividing total revenue by the number of orders made within a specific timeframe (Shopify, 2026).

### Example:

A boutique earns R80 000 from 400 orders in a month, giving a AOV of R200.

## 28. SALES FUNNEL

A model that explains the steps a customer takes, starting from discovering a business or product and ending with them making a purchase (IBM, 2024).

### Example:

A customer may first become aware of a BMW through advertising, then explore the different BMW models available and compare their options. The customer can then book a test drive before deciding to purchase the vehicle (BMW South Africa, 2026).

## Section 4

## 29. MARKETING CAMPAIGN

A planned set of marketing activities carried out over a certain period to help a business achieve a specific goal, such as increasing sales, promoting a product, or making more people aware of the brand (HubSpot, 2026).

### Example:

Coca-Cola started its "share a coke" campaign in Australia in 2011 by putting popular names on its bottles and cans. The campaign was ~~super~~ successful, with more than 250 million named bottles and cans sold in Australia.

(The Coca-Cola Company, 2014).

### 30. POST-CAMPAIGN ANALYSIS

After an email campaign, a company can track at the extra revenue generated and the ROI to see how well the campaign performed and whether the money spent on it was worthwhile (The Coca-Cola Company, 2018).

#### Example:

Coca-Cola reviewed the results of its U.S. Character Story campaign after it had ended. The company compared consumer responses before and after the campaign and found a 7% increase in trust, as well as improvements in consumption intent and brand favourability (The Coca-Cola Company, 2018).

### 31. INCREMENTAL REVENUE

The extra revenue a business earns from a specific campaign or activity, on top of what it would normally have earned without it (Shopify, 2025).

#### Example:

An online clothing store normally makes \$7000 in sales in a week. After running a promotion for winter jackets, its sales increased to \$10500. The additional \$3500 is the incremental revenue generated by the promotion (Shopify, 2025).

### 32. RETURN ON INVESTMENT

A measure used to see how much profit or return was made compared to the amount of money invested (Investopedia, 2025).

#### Example:

A campaign costing R100 000 that generates R400 000 in profit gives an ROI of  $((R400\ 000 - R100\ 000 \div 100\ 000) \times 100 = 300\%$

### 33. RETURN ON AD SPEND (ROAS)

A metric that measures how much revenue a business earns for every rand spent on advertising. (ROAS = Revenue generated from advertising  $\div$  Advertising cost) (Google Ads, 2026).

#### Example:

Atzaró Cape Town spent R155 007 on Google Ads and generated R874 532 in revenue from the campaign. This resulted in a 564% return on ad spend (Cape Marketing Agency, 2026).

### 34. CLICK THROUGH RATE

The number of people in percentage who click on an email, link or ad out of the total number of those who saw it  $(\text{clicks} \div \text{impressions} \times 100)$  (Google, 2026).

#### Example:

BrightLearn could have 5 clicks and 100 impressions, their CTR could then be 5%.

### 35. BOUNCE RATE

The percentage in which people visit the first page of a website without clicking through to any other page (semrush, n.d.)

#### Example:

A website receives 1000 sessions, and 320 of those sessions are unengaged. This gives the website a bounce rate of 32% (semrush, 2026).

### 36. A/B TESTING

The method of comparison where two versions of something (webpage, ad or email) by showing each version to a different group in order to see which one performs better on a chosen metric like CTR (Optimizely, 2026)

#### Example:

Optimizely tested two versions of its homepage by showing a dog to 50% of visitors while the other 50% saw the page without the dog.

The results showed that the visitors who saw the dog engaged with the content three times more than those who did not (Optimizely, 2026).

### 37. LEAD GENERATION

The act of attracting and capturing potential customer's interest by collecting their contact details, prior to them becoming a customer at all.

#### Example:

Daniel Nyquist, the CMO at Crosslist, introduced a

referral programme for existing customers.

Within six months, referrals made up more than 40% of the company's new business, showing how referrals can help a business generate new leads (HubSpot, 2026).

### 38. MARKETING ATTRIBUTION

The method of deciding which marketing channel should get credit for the conversion they drive especially when a customer interacted with supporting channels before purchasing (Google Analytics, 2026).

#### Example:

A customer may first see a business on social media, search for the business online, visit its website and later see another advertisement before making a purchase. It helps the business identify which of these marketing activities contributed to the customer making the purchase (Google Analytics, 2026).